

# Environmental Optimization of Food Taxation in Canada

APM348 — Mathematical Modelling

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## Abstract

Food production accounts for 23 to 42% of global human-induced greenhouse gas emissions. This study develops a constrained optimization model that recommends a system of taxes and subsidies across 14 Canadian food categories to minimize CO<sub>2</sub>-equivalent environmental impacts from household food consumption. Consumer behaviour is modeled using own and cross price elasticities following the AIDS framework using elasticity and baseline household expenditure data from Statistics Canada. This optimization model is bound by configurable political and economic constraints, including per-category tax and subsidy bounds, a cap on average household budget impact, a cap on food-sector GDP impact, and a requirement that total food intake is preserved. Outputs include optimal tax and subsidy rates, sensitivity analysis across constraints, and per-category environmental impact predictions. Under modest constraints (20% tax/subsidy cap per category, 10% overall taxation rate), the model predicts a 12.4% reduction in environmental impact (CO<sub>2</sub> equivalency) and a 3.79% reduction in average household food expenditure without compromising total household food consumption. These results suggest that, with carefully designed demand-side food taxation policies, environmental benefits can increase with minimal economic drawbacks.

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# 1 Introduction

## 1.1 Environmental Impact of Food Production

Food production has a negative environmental impact globally, accounting for 23–42% of total man-made greenhouse gas (GHG) emissions (1). Livestock in particular make up a large and disproportionate share of food-related emissions (14.5% of global GHG emissions) with beef accounting for 41% and cattle milk 20% of that figure (2). In Canada, domestic agriculture accounts for only 8% of national GHG emissions (3), but this figure understates the true environmental footprint of Canadian food consumption, which includes global supply-chain impacts such as land use change, feed production, and transportation emissions.

## 1.2 Consumer Behaviour and Price

Demand-side taxation is a policy tool for internalizing the negative externalities of consumption in a global market. When the environmental cost of a product is not reflected in its price, consumers have no market signal to prefer lower-impact alternatives. A well-designed taxation system shifts the effective price of high-emission goods upward, encouraging substitution toward lower-impact alternatives.

Price changes impact consumers' purchasing behaviors, which thereby impact the environment. Elasticity estimates show how the proportional consumption of a product responds to the proportional price change of the same, or another product (4), for example: a 10% increase in the cost of beef results in a 4.6% reduction in beef consumption and a 0.3% increase in poultry consumption. Government intervention, such as in the form of taxes and subsidies, affects prices for consumers. A tax on a food raises its market price, reducing demand and potentially driving consumers toward lower-emission substitutes. A subsidy lowers the price of a food category, incentivizing its consumption. The combination of taxes on high-emission foods and subsidies on low-emission alternatives is therefore a natural policy design, creating simultaneous pressure away from environmentally costly options and toward more sustainable ones.

## 1.3 Environmental Policy

Real-world precedents for this type of intervention exist. Denmark introduced a tax on saturated fats in meat, dairy, and oils in 2011, resulting in effective price increases of 13–16% on targeted products (11). Hungary introduced a broader “junk food” tax the same year targeting processed foods high in sugar, salt, and fat. These cases demonstrate that

food taxation at the scale considered in this study, though economically modest, pushes the boundary of what is politically achievable. Denmark’s repeal after 15 months due to industry pressure and cross-border shopping highlights that political feasibility remains a significant barrier to the implementation of any such policy.

## 1.4 Modelling Objective and Questions

Primary Question: Given constraints on maximum taxation per product, overall taxation, and maximum impact on GDP and the individual household budget, what is the optimal system of taxes and subsidies (organized by food category) to minimize the environmental impact of household food consumption, as measured across the entire supply chain?

Secondary questions addressed by this study include:

- At the optimal taxation vector, what are the binding constraints?
- How sensitive are emissions reductions to each constraint parameter?
- How do the model’s predicted results compare to real-world agriculture policy interventions?

# 2 Mathematical Approach

## 2.1 Description

The methodology in this project includes a consumer behaviour model, an environmental cost function, and an optimization algorithm. The core of the software pipeline is an optimization algorithm that optimizes for environmental impact based on the behavior predicted by the consumer behavior model, subject to the inputted constraints. The taxation/subsidy system is treated as a single vector in the space of possible systems. The space of possible taxation/subsidy systems is an  $n$ -dimensional space, where  $n$  is the number of food categories. Based on the consumer behavior model, changes to the output of the cost function are computed for possible changes in this taxation vector in each of the  $n$  basis vectors. The optimizer, which is implemented with SciPy’s `scipy.optimize` (14), uses these changes to iteratively compute the optimal taxation vector subject to the inputted constraints.

## 2.2 Consumer Behaviour Model (AIDS)

The consumer behaviour model follows the Almost Ideal Demand System developed by Deaton and Muellbauer (5), using own and cross-price elasticities and baseline consumer purchasing patterns from Statistics Canada. This choice allows a locally reliable model for how consumer purchasing behaviour varies with changes in price. In the AIDS model, average household consumer behaviour is linearly approximated, treating own and cross price elasticities as derivatives of proportional change in quantity purchased with respect to proportional change in price. The model is formalized as follows:

$$q_i(\mathbf{t}) = q_i^0 \left( 1 + \sum_j E_{ij} \cdot t_j \right)$$

where  $q_i^0$  is the baseline quantity consumed,  $E_{ij}$  is the elasticity of good  $i$  with respect to the price of good  $j$ , and  $t_j$  is the ad valorem tax rate applied to category  $j$ .

## 2.3 Objective Function

The environmental cost function is a summation over the food categories of the environmental impact of the quantity of food consumed in each category. This was formalized as:

$$\min_{\mathbf{t}} \sum_i e_i \cdot q_i(\mathbf{t})$$

where  $e_i$  is the per-unit environmental impact (measured in CO<sub>2</sub> equivalency) of food category  $i$ ,  $q_i$  is the quantity sold of category  $i$ , and  $\mathbf{t}$  is the vector of tax rates.

Data for the per-kilogram impact of products was sourced from Our World in Data (7), which offers a carbon equivalency calculation that accounts for a range of agriculture-driven environmental impacts.

## 2.4 Constraints

The optimization is subject to five economic and political constraints:

1. Each food category's tax rate must not exceed 20%. Each food category's subsidy rate must not exceed 20% (equivalent to  $-20\%$  tax rate).
2. The budget-share-weighted average tax across all food categories cannot exceed 10%. This limits the overall tax burden and prevents the optimizer from applying maximum

tax rates across all categories simultaneously.

3. Household food expenditure must not increase by more than 5%. This ensures grocery bills do not increase sharply, as food is a necessity.
4. Food-sector GDP must stay above 95% of baseline. This constraint protects the domestic food economy; if consumer spending on food drops too sharply, it harms participants across the food supply chain.
5. Total kilograms consumed must be preserved at baseline levels, ensuring the policy shifts dietary composition rather than reducing total food intake — the goal is to change what people eat, not how much they eat.

## 2.5 Optimization Method

The optimizer is implemented using SciPy’s `scipy.optimize.minimize` with the Sequential Least-Squares Quadratic Programming (SLSQP) method (14). SLSQP works well with this modeling problem, which involves a relatively small taxation vector (14-dimensional) and a smooth cost function based on polynomial approximations, the recommended use case. The algorithm begins from a zero tax vector representing the baseline state. At each iteration, SLSQP solves a small quadratic subproblem: it locally approximates the objective as a quadratic bowl and each constraint as a linear boundary, finds the optimal step within that local approximation, then verifies that the step reduces the true objective before continuing. The Hessian is also updated at each stage. Gradients are estimated by finite differences, perturbing each of the 14 tax rates slightly and measuring the resulting change in emissions. The algorithm terminates when the step size becomes negligibly small, indicating a local minimum, or when no feasible improving direction exists, indicating that the solution is bounded by one or more active constraints. To verify that the solution is a global rather than local optimum, the optimizer was run from multiple random starting points in addition to the zero initialization. All runs converged to the same objective value, providing strong evidence that the result is the true global optimum within the feasible region.

## 3 Data

### 3.1 Own and Cross-Price Elasticity

The  $14 \times 14$  price elasticity matrix was sourced from Agriculture and Agri-Food Canada (2006) (4). Entry  $E_{ij}$  represents the proportional change in quantity demanded of food

category  $i$  in response to a proportional change in the price of food category  $j$ . Diagonal entries (own-price elasticities) are negative, indicating that a price increase reduces consumption of that good. Off-diagonal entries (cross-price elasticities) can be positive (substitutes) or negative (complements). One entry in the original AAFC data contained a suspected typographical error: the (Pork, Vegetable) cross-price elasticity was recorded as  $-010368$ , which is inconsistent with the scale of all other entries in the matrix. This was corrected to  $-0.10368$ . The full matrix is presented in Table 1 below.

Table 1: Price Elasticity Matrix (Source: AAFC, 2006)

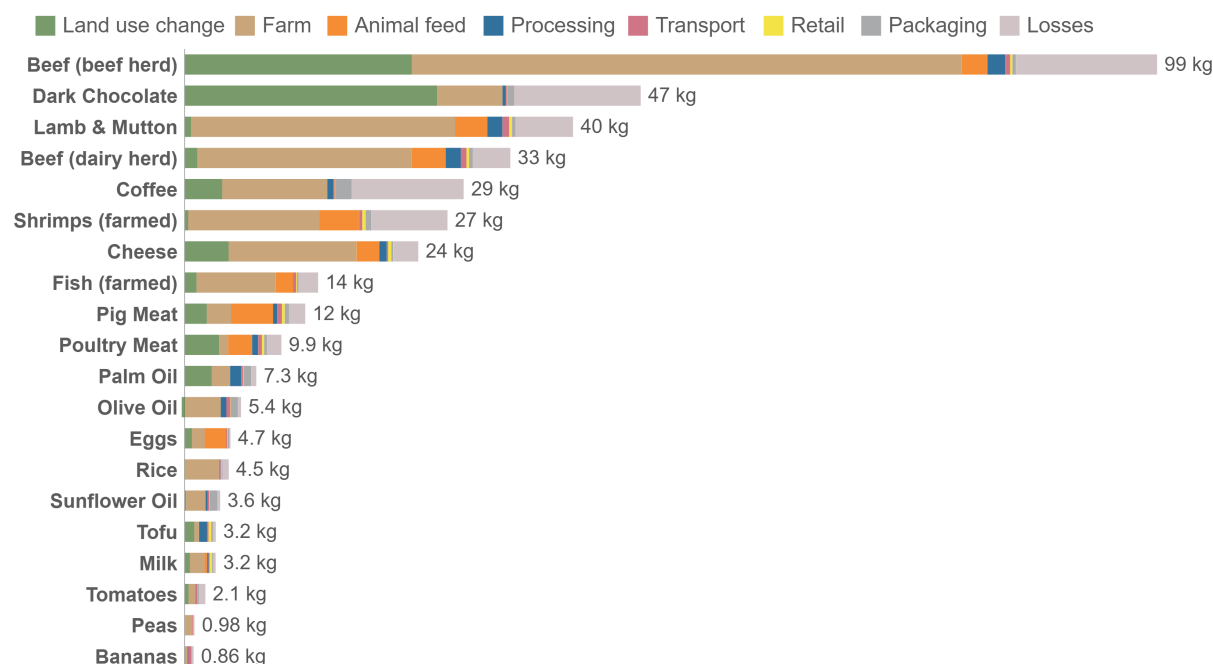
|               | Beef     | Poultry  | Pork     | Other meat | Fish     | Dairy    | Eggs     | Bakery   | Cereal   | Fruit    | Vegetable | Fats & oils | Non-alc. bev. | Other food |
|---------------|----------|----------|----------|------------|----------|----------|----------|----------|----------|----------|-----------|-------------|---------------|------------|
| Beef          | -0.45615 | 0.02531  | 0.02640  | -0.06541   | -0.06081 | -0.03087 | -0.00168 | -0.07484 | -0.05300 | -0.01592 | -0.08208  | -0.01419    | -0.00667      | -0.02859   |
| Poultry       | 0.02914  | -0.81954 | -0.00057 | 0.00334    | -0.05686 | 0.03598  | -0.02079 | -0.06838 | 0.01113  | -0.00526 | -0.01961  | 0.00028     | 0.03366       | 0.03553    |
| Pork          | 0.05620  | 0.00162  | -0.68006 | -0.06700   | -0.04931 | 0.03826  | 0.01155  | -0.04393 | 0.06836  | -0.00362 | -0.10368  | -0.01445    | 0.00873       | -0.02384   |
| Other meat    | -0.06849 | -0.00079 | -0.03599 | -0.80841   | -0.02524 | 0.00710  | -0.00423 | -0.07585 | 0.06925  | 0.05925  | 0.00083   | -0.02956    | 0.00441       | 0.00497    |
| Fish          | -0.10042 | -0.08044 | -0.04129 | -0.03605   | -0.10529 | 0.00296  | -0.01469 | -0.10698 | -0.16800 | -0.08109 | -0.01015  | -0.02614    | 0.03258       | -0.05898   |
| Dairy         | -0.03679 | 0.00323  | 0.00103  | -0.00966   | -0.01156 | -0.88486 | -0.01450 | -0.04092 | -0.04218 | -0.03593 | -0.02436  | -0.01374    | 0.01351       | 0.01415    |
| Eggs          | 0.00879  | -0.07187 | 0.03463  | 0.00092    | -0.03354 | -0.05829 | -0.35029 | -0.09091 | -0.07972 | -0.01761 | -0.09597  | 0.00799     | 0.03044       | 0.10410    |
| Bakery        | -0.07403 | -0.05858 | -0.02441 | -0.07199   | -0.06206 | -0.04273 | -0.02173 | -0.43027 | -0.02191 | -0.01754 | -0.12457  | -0.02232    | -0.02718      | 0.05829    |
| Cereal        | -0.07092 | 0.00870  | 0.03910  | 0.08935    | -0.12841 | -0.06489 | -0.02509 | -0.02719 | -0.70802 | -0.02209 | -0.08680  | -0.01933    | 0.06075       | 0.06037    |
| Fruit         | -0.04473 | -0.03164 | -0.01907 | 0.01449    | -0.05585 | -0.06811 | -0.01298 | -0.04313 | -0.03576 | -0.84626 | -0.06308  | -0.01866    | 0.00922       | -0.06844   |
| Vegetable     | -0.10236 | -0.04376 | -0.06071 | -0.03050   | -0.02734 | -0.06105 | -0.02698 | -0.14608 | -0.08104 | -0.07392 | -0.65117  | -0.03662    | 0.00584       | 0.02767    |
| Fats & oils   | -0.04505 | 0.01092  | -0.02416 | -0.10388   | -0.05621 | -0.04165 | 0.00540  | -0.07862 | -0.04871 | -0.04089 | -0.12090  | -0.22675    | 0.02254       | 0.00159    |
| Non-alc. bev. | -0.01218 | 0.03051  | 0.00154  | 0.00538    | 0.01794  | 0.04930  | 0.00299  | -0.03228 | 0.05652  | 0.05476  | 0.04568   | 0.01860     | -1.14449      | 0.00886    |
| Other food    | -0.03085 | 0.00169  | -0.01542 | -0.00718   | -0.02907 | 0.01500  | 0.00404  | 0.02356  | 0.01468  | -0.02468 | 0.03984   | -0.00598    | -0.00638      | -1.03697   |

## 3.2 CO<sub>2</sub> Emissions Data

### Food: greenhouse gas emissions across the supply chain



Greenhouse gas emissions<sup>1</sup> are measured in kilograms of carbon dioxide-equivalents (CO<sub>2</sub>eq)<sup>2</sup> per kilogram of food.



Data source: Joseph Poore and Thomas Nemecek (2018).

OurWorldinData.org/environmental-impacts-of-food | CC BY

**1. Greenhouse gas emissions** A greenhouse gas (GHG) is a gas that causes the atmosphere to warm by absorbing and emitting radiant energy. Greenhouse gases absorb radiation that is radiated by Earth, preventing this heat from escaping to space. Carbon dioxide (CO<sub>2</sub>) is the most well-known greenhouse gas, but there are others including methane, nitrous oxide, and in fact, water vapor. Human-made emissions of greenhouse gases from fossil fuels, industry, and agriculture are the leading cause of global climate change. Greenhouse gas emissions measure the total amount of all greenhouse gases that are emitted. These are often quantified in carbon dioxide equivalents (CO<sub>2</sub>eq) which take account of the amount of warming that each molecule of different gases creates.

**2. Carbon dioxide equivalents (CO<sub>2</sub>eq)** Carbon dioxide is the most important greenhouse gas, but not the only one. To capture all greenhouse gas emissions, researchers express them in "carbon dioxide equivalents" (CO<sub>2</sub>eq). This takes all greenhouse gases into account, not just CO<sub>2</sub>. To express all greenhouse gases in carbon dioxide equivalents (CO<sub>2</sub>eq), each one is weighted by its [global warming potential \(GWP\) value](#). GWP measures the amount of warming a gas creates compared to CO<sub>2</sub>. CO<sub>2</sub> is given a GWP value of one. If a gas had a GWP of 10 then one kilogram of that gas would generate ten times the warming effect as one kilogram of CO<sub>2</sub>. Carbon dioxide equivalents are calculated for each gas by multiplying the mass of emissions of a specific greenhouse gas by its GWP factor. This warming can be stated over different timescales. To calculate CO<sub>2</sub>eq over 100 years, we'd multiply each gas by its GWP over a 100-year timescale (GWP100). Total greenhouse gas emissions – measured in CO<sub>2</sub>eq – are then calculated by summing each gas' CO<sub>2</sub>eq value.

Figure 1: Per-Category GHG Emissions Equivalencies Chart

The per-unit CO<sub>2</sub> emissions equivalencies data comes from the dataset compiled by Ritchie, Rosado, and Roser on Our World in Data (7), based on the meta-analysis by Poore and Nemecek (6). This dataset reports environmental impact in kilograms of CO<sub>2</sub> emissions equivalent to the overall environmental impact per kilogram of product for 43 individual food items. Since our model consists of 14 food categories, we mapped each of the 43 items

to its corresponding category.

For some categories: Beef (99.5 kg CO<sub>2</sub>-eq/kg), Poultry (9.9 kg CO<sub>2</sub>-eq/kg), Pork (12.3 kg CO<sub>2</sub>-eq/kg), Other meat from Lamb and Mutton (39.7 kg CO<sub>2</sub>-eq/kg), and Eggs (4.7 kg CO<sub>2</sub>-eq/kg), the mapping is one-to-one between the two sources.

For food categories consisting of multiple sub-categories, we computed an unweighted average within each category. Dairy (20.11 kg CO<sub>2</sub>-eq/kg) averages Beef (dairy herd), Cheese, and Milk, with the dairy herd entry fully assigned to Dairy rather than Beef because dairy cattle are bred for milk production. Non-alcoholic beverages (14.75 kg CO<sub>2</sub>-eq/kg) averages only Coffee and Soy milk; this is the least representative category in our model since common items such as juices, teas, and soft drinks are absent from the source dataset, and the average is heavily skewed by coffee’s high emissions. Fruit (0.85 kg CO<sub>2</sub>-eq/kg) averages five items, Vegetable (0.85 kg CO<sub>2</sub>-eq/kg) averages eight items, Cereal (2.45 kg CO<sub>2</sub>-eq/kg) averages four items, Fats and oils (5.29 kg CO<sub>2</sub>-eq/kg) averages five oils, and Other food (7.76 kg CO<sub>2</sub>-eq/kg) averages eight items including sugars, dark chocolate, nuts, pulses, tofu, and wine. Full calculations and breakdowns for each category are provided in Appendix A.

### 3.3 Baseline Household Food Expenditure

The baseline household food expenditure data was sourced from Statistics Canada Table 11-10-0125-01 (8), which consists of average annual food spending per Canadian household across eight food categories for food purchased from stores, totalling \$5,709. We selected 2010 expenditure data for its temporal proximity to the 2006 elasticity estimation period, ensuring consistency between the baseline budget shares and the consumer behaviour patterns captured by the elasticity matrix. However, our elasticity matrix from Agriculture and Agri-Food Canada (4) was defined across 14 food categories, requiring a mapping between the two classifications.

Four categories map directly without need for splitting: Bakery products (\$581), Cereal grains and cereal products (\$359), Fish and seafood (\$199), and Vegetables and vegetable preparations (\$616).

The remaining four Statistics Canada categories each contain multiple categories from our model and are split using estimated proportions derived from Canadian consumption data. We divided Meat (\$1,027) into Beef, Poultry, Pork, and Other meat, with estimated expenditure splits of 35%, 27%, 23%, and 15% respectively, based on approximate per-capita consumption volumes from the Statistics Canada Food Statistics report (9) and average retail price differences across meat categories. Similarly, Dairy products and eggs (\$899) were

split into Dairy and Eggs at an estimated 89% and 11% respectively. Fruit, fruit preparations and nuts (\$713) was split into Fruit at 92% and Other food at 8% to account for the nuts portion. Finally, Non-alcoholic beverages and other food products (\$1,315) were divided into Non-alcoholic beverages at 28%, Fats and oils at 8%, and Other food at 64%.

## 4 Numerical Simulations and Results

Based on our constraints and assumptions, we ran simulations to identify optimal tax rates for each food category that would reduce food consumption, thereby lowering overall carbon emissions. Overall, we saw a 12.4% reduction in carbon emissions.

### 4.1 Optimal Tax and Subsidy Rates

With the constraints fixed as described in section 2.4 the categories that were taxed at the maximum rate included beef, non-alcoholic beverages, and dairy, all of which have high elasticities, budget shares, and per-kilogram CO<sub>2</sub> (eq). The products with low emissions were mainly “negatively” taxed, allowing the optimizer to tax other foods heavily without violating the overall food consumption constraint. Without this constraint most food categories were taxed heavily and overall food consumption dropped by 9%.

Figure 2 shows the optimal tax rates for each food category when setting the maximum tax rate to 20% (and likewise the maximum subsidy of -20%). Figure 3 shows the optimal tax rates at different maximum per-product tax constraints, with all other constraints held fixed.

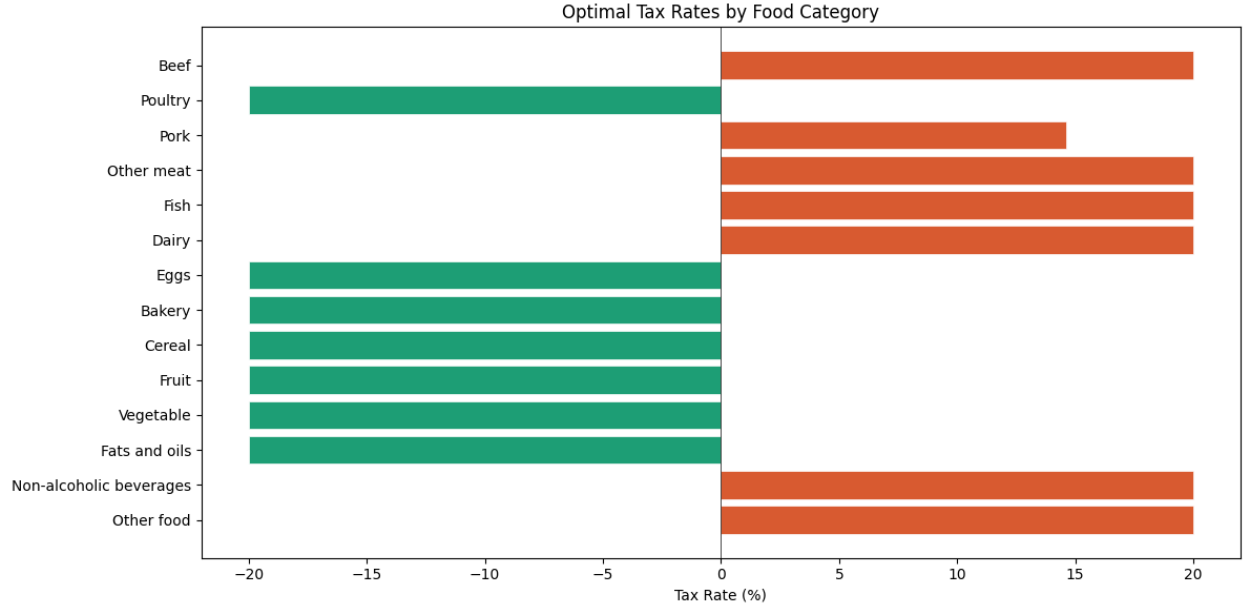


Figure 2: Optimal Tax Rates by Food Category Chart



Figure 3: Optimal Tax Rates vs Max Tax Per Product Heatmap

## 4.2 Consumption Changes

Results were obtained that showed changes in consumption for each food category under the optimized tax rates. We can see from Figure 4 that consumers ended up choosing available substitutes for the categories that got heavily taxed, as some categories went up in consumption while others reduced. For example, beef consumption was reduced by 10%

while poultry consumption increased by 20%, which was the highest increase out of all the foods, suggesting that consumers would choose poultry as a substitute for beef. Other largely consumed and high impact categories, such as dairy and non-alcoholic beverages, showed a significant reduction in consumption.

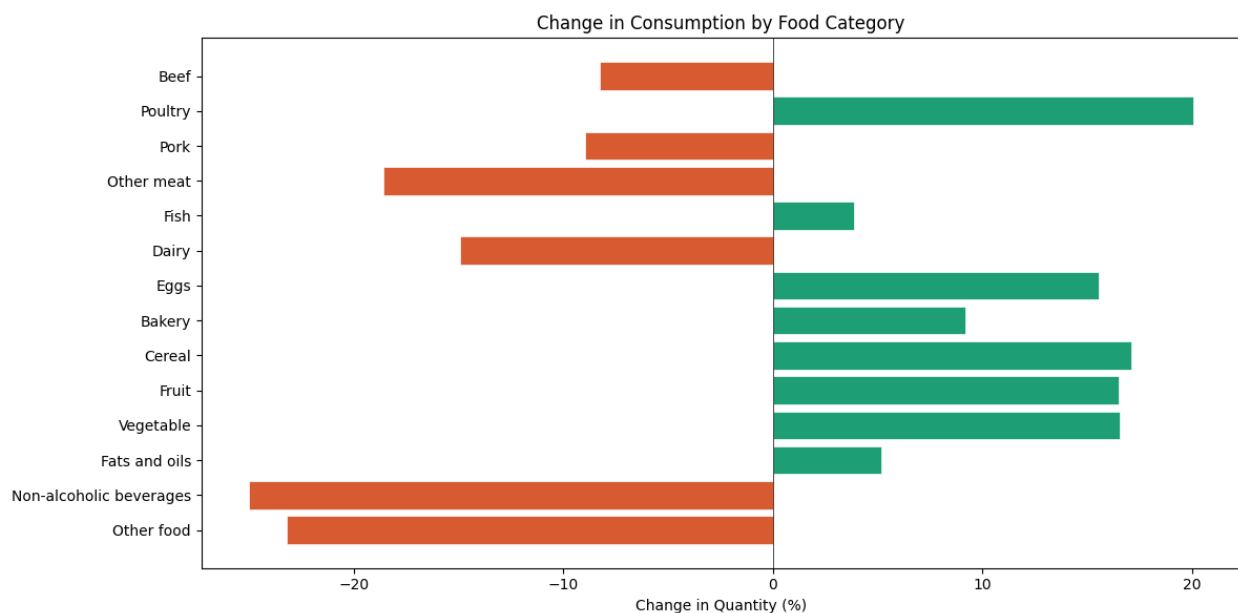


Figure 4: Change in Consumption by Food Category Chart

### 4.3 Emissions Reduction

Food categories with high per-kilogram environmental impacts, high budget shares, and high elasticities were taxed heavily as expected, resulting in a decreased overall environmental impact. We can see the baseline versus optimized levels of carbon emissions for each category, reflecting an overall 12.4% reduction in CO<sub>2</sub> (eq). For example, dairy, which initially showed a high level of carbon emissions at approximately 3500 kg CO<sub>2</sub> (eq), reduced to about 3000 kg.

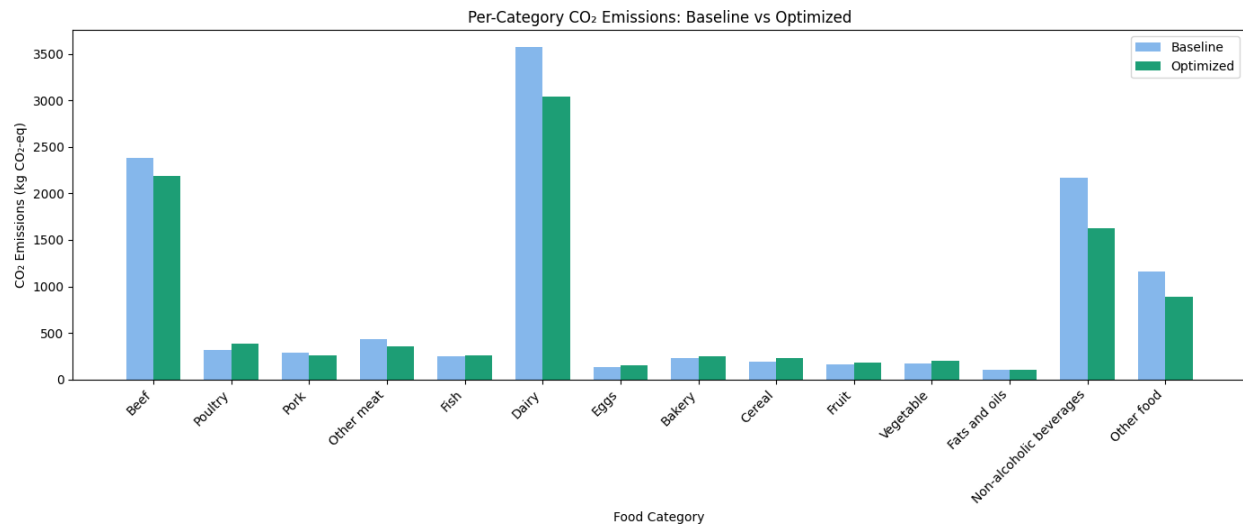


Figure 5: Per-Category CO<sub>2</sub> Emissions Chart: Baseline vs Optimized

#### 4.4 Politically Ambitious Constraints

Further simulations were performed, adjusting constraints to a more politically ambitious level. The maximum per-category taxation/subsidy rate was set at 50%, and the maximum food sector GDP impact was set at 15%, holding other constraints fixed. In this more ambitious model, emissions from dairy consumption reduced to about 2200 kg CO<sub>2</sub> (eq), as seen in Figure 6. Similarly, stronger reductions were observed in the beef, other food, and non-alcoholic beverages categories. Overall, a 29.91% CO<sub>2</sub>(eq) reduction was observed.

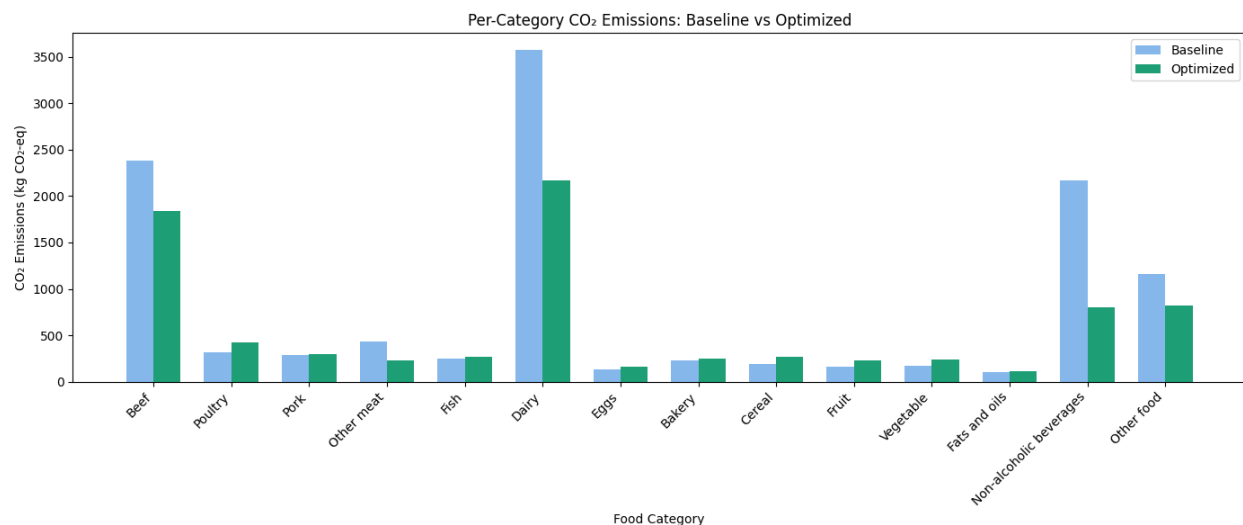


Figure 6: Per-Category CO<sub>2</sub> Emissions Chart: Baseline vs Optimized (Loosened)

## 4.5 Sensitivity Analysis

A sensitivity analysis was performed on the original model to show how sensitive the output is to each parameter. Based on the sensitivity graphs (Figures 7 - 9), we observe that the output is highly sensitive to the maximum tax per product and the total tax received and not sensitive to the maximum impact the budget can take as seen by the approximate first derivatives of these curves at their boundary points. It is noteworthy that the maximal budget impact constraint is non-binding. Due to the constraint that overall food consumption must be held at baseline, the foods that consumers are inclined to use as substitutes for heavily taxed items are themselves heavily subsidized. The result is that the average household spending actually decreases by 3.79%.

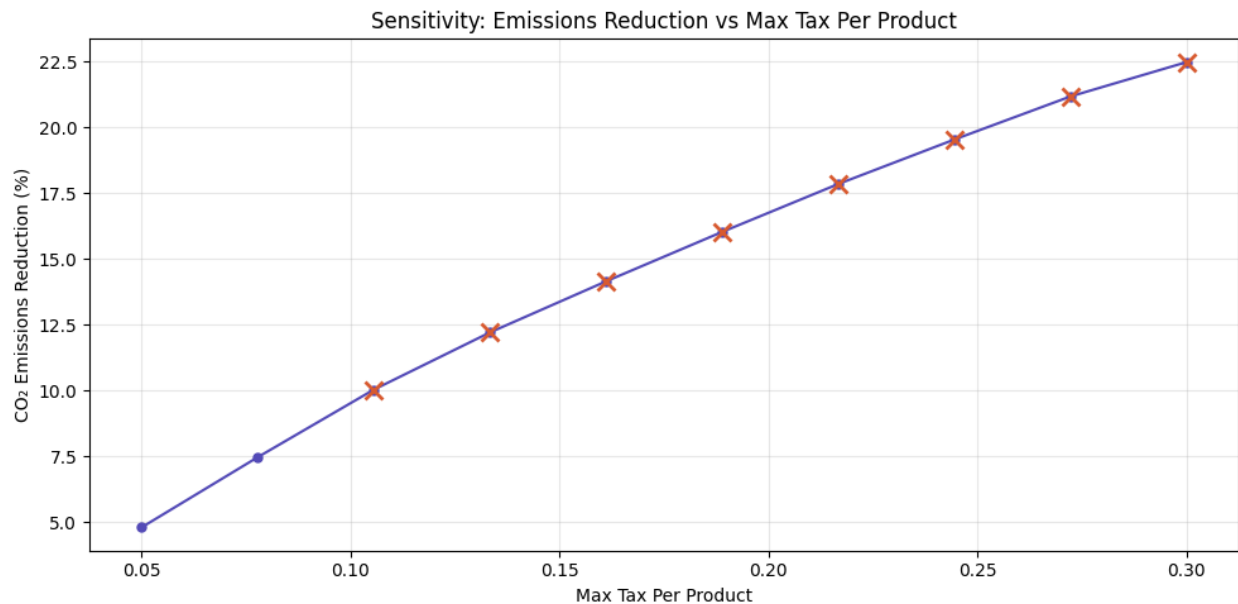


Figure 7: Sensitivity Graph: Emissions Reduction vs Max Tax Per Product

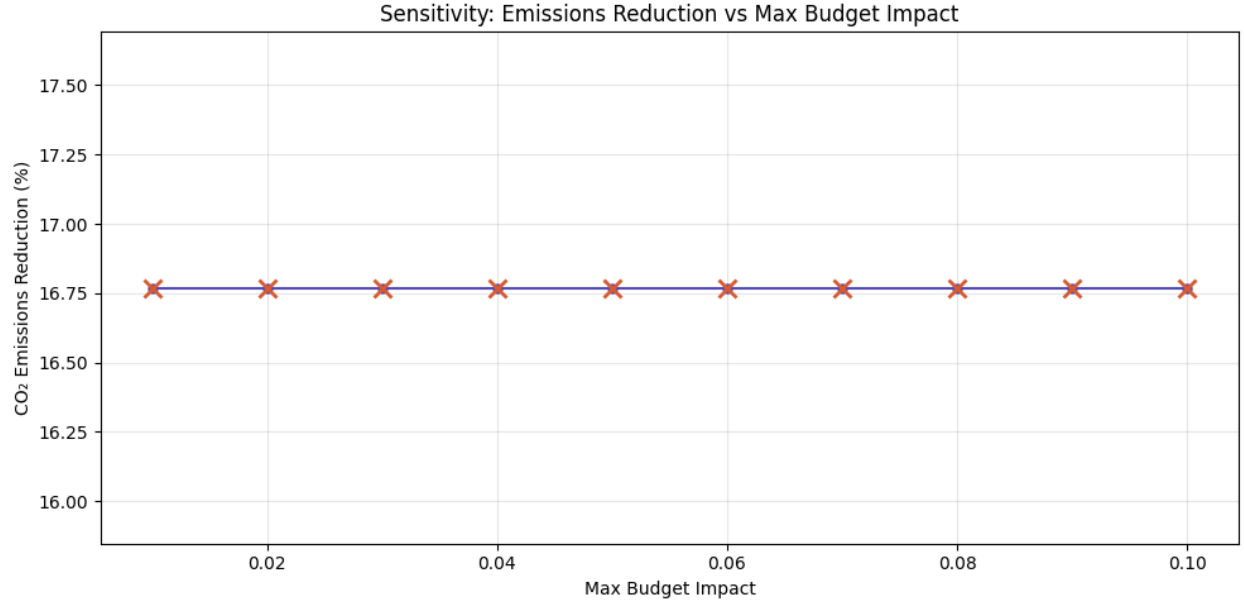


Figure 8: Sensitivity Graph: Emissions Reduction vs Max Budget Impact

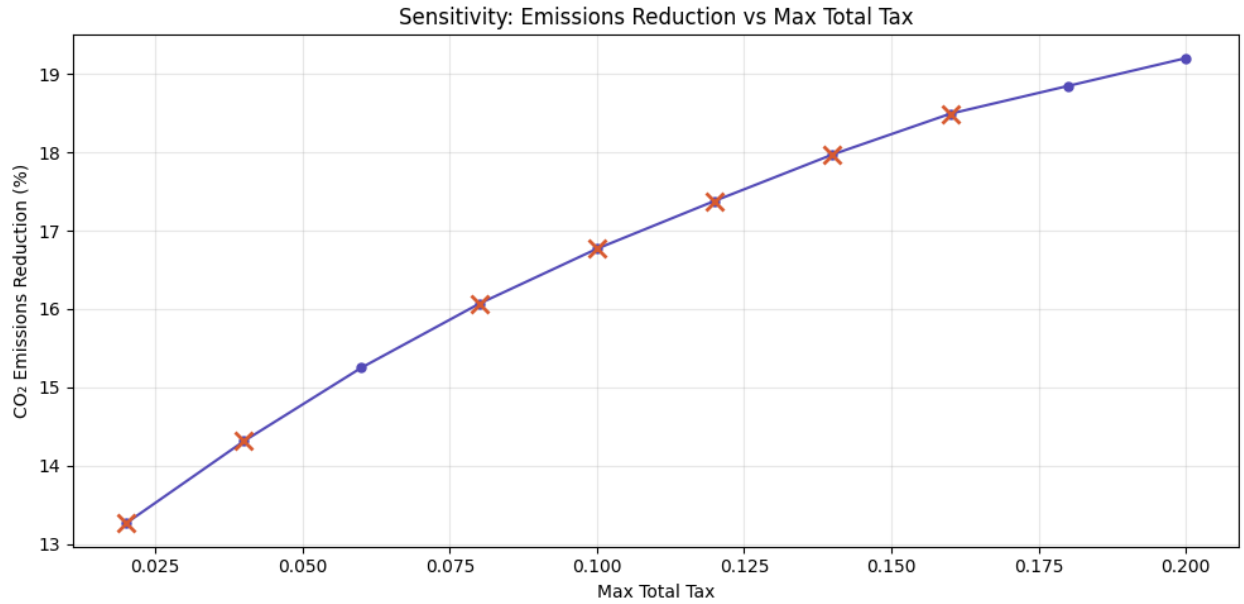


Figure 9: Sensitivity Graph: Emissions Reduction vs Max Total Tax

At the optimal taxation vector the binding constraints were the caps on taxation levels, indicating that the optimizer was bound by the 10% aggregate tax cap and the 20% per-category tax cap. Table 2 shows sensitivity analyses for the constraints pictured in figures 7 - 9. Relaxing the per-product cap on taxation by 1% yields an additional 0.66% of emissions reductions, while relaxing the total tax cap by 1% yields only 0.33% emissions reductions.

The household budget and food-sector GDP constraints both had significant slack, meaning the optimal policy satisfied it comfortably without it influencing the solution. The primary policy lever, therefore, is a higher level of taxation on food categories with high per-kilogram emissions, high budget shares, and high substitution effects.

| <b>Constraint</b>       | <b>Local Sensitivity (pp / 1pp relaxation)</b> | <b>Binding?</b> |
|-------------------------|------------------------------------------------|-----------------|
| Per-product tax cap     | 0.66                                           | Yes             |
| Weighted total tax cap  | 0.33                                           | Yes             |
| Household budget impact | 0.00                                           | No              |

Table 2: Local sensitivity of emissions reduction to each constraint parameter at the optimized tax vector (20% per-product cap, 10% total tax cap, 5% budget impact cap).

## 5 Contributions

### 5.1 Ben Lemke

- Contributed to Mathematical Approach section, Introduction section, Conclusion section, and Limitations Section
- Presented methodology section and answered questions to a live audience
- Contributed to code repository
- Developed cost function and AIDS implementation

### 5.2 Medha Neti

- Contributed to Results section, Question and answer section, and Future Work section
- Presented results and conclusions sections and answered questions to a live audience
- Contributed to Google Colab simulations
- Performed sensitivity analysis

### 5.3 Aaron Ren

- Contributed to Data section, Conclusion section, and Introduction section
- Presented background section and answered questions to a live audience
- Contributed to code repository and simulations

- Generated plots and researched background material and data
- Performed data aggregation

Special thanks to Noah Frankl for his insightful discussions of economics and political background

## 6 Responses to Presentation Questions

**Question:** How would your model change if transportation emissions were taken into consideration?

**Response:** The impact of transportation was included in the CO<sub>2</sub> equivalency data sourced from Our World in Data (7), and so is taken into consideration by the cost function of the model. It is worth noting that low-emission foods have the highest transport emissions share (Beet Sugar at 34.7%, Bananas at 34.3%, and root vegetables at 26.7%) while the high-emission foods have the lowest (Beef at 0.5%, Dark Chocolate at 0.2%, Coffee at 0.5%, Cheese at 0.6%, and Lamb at 1.7%).

**Question:** How were your constraints calculated? What other parameters could be valuable for minimizing CO<sub>2</sub> emissions?

**Response:** Constraints for the various simulations run were determined in discussion with colleagues studying political science and economics. The taxation constraints align with real-world case studies, namely the policies imposed by the administrations of Denmark and Hungary in 2011. Politically, the 20% tax/subsidy cap per category is ambitious, but economically, the constraints for the simulations done are chosen to be quite modest, allowing at most a 5% impact on food-sector GDP. Ultimately, the primary contribution of this project is not the particular taxation rates or choice of constraints, but a systematic method for optimizing the impact of environmental policy under any set of constraints as is determined by the public discourse in a given place and time.

**Question:** Do you think the 12.4% reduction in CO<sub>2</sub> emissions is a hard limit, or could it be improved by refining or adding assumptions?

**Response:** The accuracy of the model's outputs could be improved by accounting for differences in consumers based on income, region, and cultural factors. As described by the sensitivity analyses a refinement or shift of constraint parameter values would have a significant impact on results. The 12.4% figure is highly contingent on these modelling decisions, and would be expected to vary with an improvement to the demand model or the

refinement of the constraints.

**Question:** How are fats and oils defined? Aren't they correlated with beef or dairy?

**Response:** The Statistics Canada sources used do not contain strict definitions for categories like fats and oils. They do however indicate that the relevant category would include margarine and salad oils, indicating that a crossover with meat, dairy, and egg categories was not intended. For the purposes of this study each food category was taken to be completely distinct.

**Question:** What suggestions can you make to mitigate the short-horizon limitation?

**Response:** While it is true that this model assumes that consumer behaviour shifts instantaneously, this is not a major limitation of the study. Such an assumption is a reasonable approximation which greatly reduces computational complexity. In real-world applications administrations are able to measure the impact on consumer behaviour and therefore observe the time required for behaviour to respond to prices before adjusting the taxation system based on observations.

**Question:** When you mentioned that the amount of food per person would be constant, were you referring to caloric intake or mass of food? What is the impact of this assumption on consumer choices and, consequently, emissions?

**Response:** The constraint that the total quantity of food (measured by mass) is constant is not an assumption of the demand model at all, but merely a constraint for the purposes of the optimizer. This means that the optimizer is bounded by total food consumption. In simulations done without this constraint, taxes were increased on most food items so that overall food consumption decreased dramatically—an unsustainable policy recommendation.

**Question:** Your analysis focuses on CO<sub>2</sub> emissions, but environmental impact includes many other factors. How do you account for non-carbon environmental impacts, and could minimizing CO<sub>2</sub> emissions lead to unintended negative effects in other areas?

**Response:** The model uses a metric for environmental impact from Our World in Data (7) which quantifies a range of carbon and non-carbon impacts and assesses their carbon equivalency, referred to throughout the report and presentation as CO<sub>2</sub> (eq). Therefore, the results minimize overall environmental impact, not only carbon emissions. Importantly, many environmental impacts are dependent on each other in a way that if one sustainability

metric improves, another will as well. This aspect is of course far beyond the scope of this project. However, as we included in our section about future work, one could look at other environmental metrics to improve the model, such as water and land usage. Keeping track of the water footprints of each food category and constraining how much land is used could provide more precise tools for policymakers depending on their political context.

**Question: There is a hard cap of 20% tax, uniform tax rates across food items, and even the possibility of negative tax rates. How were these tax policies determined, how realistic are they in practice, and how do you balance maximizing effectiveness with policy feasibility and fairness?**

**Response:** The tax and subsidy bounds at 20% reflect real-world policy precedents. For example, European Value Added Tax (VAT) rates on food reach up to 25% in Denmark. Although the optimal tax/subsidy rates of some food categories are equivalent, the tax rates are not all uniform. Negative tax rates represent subsidies, which is a standard component of European VAT proposals. These constraints are configurable parameters, which allow policymakers to adjust the bounds based on their political and economical context.

**Question: What would you do with the tax revenue generated?**

**Response:** The redistribution or allocation of tax revenue is far beyond the scope of this project. Possible suggestions include a rebate as was implemented in Canada during the application of a consumer-facing carbon tax, and reallocation to support retraining and retrofitting efforts for workers and farm owners whose livelihoods are impacted.

**Question: Can increased tax revenue be used to compensate for additional CO<sub>2</sub> emissions through carbon capture or tree plantation subsidies?**

**Response:** The impact of carbon capture technology and tree plantation were not assessed during this study. However, creative allocation of tax revenue towards other environmental projects is clearly a sensible option for policymakers to consider, and would constitute a further environmental benefit to the policies recommended by this model.

**Question: What additional factors could affect the consumption and import/export of different foods?**

**Response:** Aside from the factors mentioned in the limitations section, body composition, health, age, and individual nutritional needs would all impact a person's consumption patterns. Import/export patterns are beyond the scope of this study. As discussed in the future work section it is reasonable to expect that the appropriate policy interventions vary by region. Even within Canada, inter-provincial import/export patterns could be included.

**Question:** What policies could reduce overall environmental impact?

**Response:** The model tends to recommend taxation on high-impact foods that have large baseline consumptions and large own-price elasticities. More broadly, policies beyond taxation such as subsidizing sustainable farming practices, investing in alternative proteins, implementing emissions labelling, or regulating production methods could complement demand-side interventions.

**Question:** What exactly does a negative tax rate mean?

**Response:** We are describing subsidies offered to retailers selling products from certain food categories as ‘negative taxes’. This is because the AIDS model does not differentiate between sources of a price change.

**Question:** These taxes may disproportionately affect low-income households, potentially limiting their access to sufficient nutrition while wealthier households remain unaffected. How would you address this issue?

**Response:** This project’s use of a single representative household is certainly a limitation, and consumers from different income brackets would definitely be expected to respond differently to price changes. However, our model actually predicts a  $-3.79\%$  impact on the average household expenditure at the 20% per-category tax cap simulated. In other words, the average household is expected to spend less money on groceries, not more, and this impact would be expected to be exaggerated in the case of low-income consumers who are not able to afford expensive options. A genuine downside for such consumers could be their reduced consumption of presumed-desirable food categories. Whether reduced consumption of these categories has negative nutritional consequences is beyond the scope of this study and would require a separate nutritional analysis.

## 7 Conclusion

### 7.1 Summary of Findings

Per-product taxation limits were the primary constraint on the optimal environmental impact reduction of demand-side taxation policies. Under a 20% cap, the model achieved a 12.4% reduction in CO<sub>2</sub> (eq) emissions. Total food intake was maintained at baseline levels, while a large substitution effect was predicted. This was reflected in meat substitutions, for example poultry consumption was predicted to increase by 20% while beef consumption fell by 10%.

The sensitivity analysis showed that the per-product tax cap and the weighted total tax

constraint were the primary drivers of the result, while the household budget constraint was never binding. Under the optimal policy, households spent 3.79% less on food overall, suggesting that well-designed tax-and-subsidy combinations need not directly overburden individual household budgets. This finding has direct policy relevance: the political concern that food taxes are regressive may be mitigated when subsidies on low-emission staples such as vegetables, fruit, and cereal are included as part of the policy package.

## 7.2 Assessment of the Mathematical Models

The primary strength of our model is that consumer behavior is determined by empirically estimated elasticities, with no additional assumptions about total expenditure or utility maximization. However, the model is a first-order linear approximation and could plausibly lose accuracy for large price changes. The predicted 12.4% reduction is consistent with the range of outcomes documented in real-world interventions. Denmark’s 2011 saturated fat tax produced price increases of 13–16% before its repeal (11), and European modeling studies estimating VAT reforms on meat and dairy report emissions reductions in the 3.5–5.7% range (10; 13). The 12.4% figure predicted here exceeds these estimates, which is plausible given that our model optimizes across all 14 categories simultaneously rather than targeting a single food category, but this comparison also suggests the result should be interpreted as an upper bound.

## 7.3 Limitations

Limitations of this study include a temporal and slight categorical mismatch of data, a simplified treatment of nutritional equivalence, and the use of a single representative household. Due to availability of data, own and cross price elasticities were used from 2006, while baseline consumer spending was used from 2010. The food category split proportions are estimates based on approximate per-capita consumption volumes and average retail price differences instead of a single breakdown. Future work could improve precision by using more detailed expenditure breakdown data from 2010. Nutritional factors were largely suppressed in this study, except as captured by cross price elasticities. The behavioral constraint implemented was exclusively a function of the mass of food consumed (measured in kg). A single representative household was assumed, thereby glossing over the impacts of income level, location, and cultural factors on consumer behavior. This study suffers also from the limitation that policies of the type recommended are likely to face significant push-back even in their most modest form from local agriculture industry, and therefore suffer from a lack of political feasibility. The development of political conditions that allow for such reform is

a prerequisite for implementation, but is beyond the scope of this study.

## 7.4 Suggestions for Future Investigations

1. One future extension of this work could be to consider consumers in other countries, other than just Canada. Then our results can either be applicable on a global scale, or we can decide constraints individually for each region. As a result, we would need to focus on other factors that affect consumer demand, such as the availability of substitutes and the proportion of income consumers spend on different goods, as these vary per region, causing consumer demand and price elasticities to be different. Utilizing the QUAIDS model would be beneficial for this, as it could provide us with consumer elasticities that take into account different income categories. However, not just economic, but also political and cultural factors, such as religious dietary preferences, should be taken into account. As a result of more comprehensive research, we would be able to better understand how all these factors depend on each other.
2. Secondly, apart from carbon emissions, we can look at other sustainability measures that are also affected, such as water use, land use, and impacts on species and habitats, and also evaluate how significant these impacts are to be accounted for (which could vary per region). For example, water usage/footprint is almost, if not equally, as important as carbon emissions, and while the carbon footprint may be minimized for a food item, the water footprint could still be high, so this impact must be accounted for and measured separately. Of course, metrics depend on the region as well, since different regions have varying sustainability requirements and resource availability. Considering these in the future would help move towards global-scale sustainability.
3. A last future work could be to consider other environmental policies affecting food production, which may also have fewer economic drawbacks than taxation policies, and improve our model based on these policies. These could include subsidizing sustainable farming practices, investing in technological innovation, and regulating production methods. We would be able to identify which policies are economically effective in regulating food production.

## A Emissions Aggregation Calculations

The following calculations show the unweighted averages used to derive per-category CO<sub>2</sub> emissions from the 43-item Poore and Nemecek dataset (6). All values are in kg CO<sub>2</sub>-eq/kg of product.

### **Dairy:**

$$(\text{Beef dairy herd (33.30)} + \text{Cheese (23.88)} + \text{Milk (3.15)}) / 3 = 60.33 / 3 = 20.11$$

### **Non-alcoholic beverages:**

$$(\text{Coffee (28.53)} + \text{Soy milk (0.98)}) / 2 = 29.51 / 2 = 14.75$$

### **Fruit:**

$$(\text{Apples (0.43)} + \text{Bananas (0.86)} + \text{Berries \& Grapes (1.53)} + \text{Citrus Fruit (0.39)} + \text{Other Fruit (1.05)}) / 5 = 4.26 / 5 = 0.85$$

### **Vegetable:**

$$(\text{Brassicas (0.52)} + \text{Cassava (1.32)} + \text{Onions \& Leeks (0.50)} + \text{Other Vegetables (0.53)} + \text{Peas (0.97)} + \text{Potatoes (0.46)} + \text{Root Vegetables (0.43)} + \text{Tomatoes (2.09)}) / 8 = 6.82 / 8 = 0.85$$

### **Cereal:**

$$(\text{Barley (1.18)} + \text{Maize (1.70)} + \text{Oatmeal (2.48)} + \text{Rice (4.45)}) / 4 = 9.81 / 4 = 2.45$$

### **Fats and oils:**

$$(\text{Olive Oil (5.43)} + \text{Palm Oil (7.32)} + \text{Rapeseed Oil (3.77)} + \text{Soybean Oil (6.32)} + \text{Sunflower Oil (3.60)}) / 5 = 26.44 / 5 = 5.29$$

### **Other food:**

$$(\text{Beet Sugar (1.81)} + \text{Cane Sugar (3.20)} + \text{Dark Chocolate (46.65)} + \text{Groundnuts (3.23)} + \text{Nuts (0.43)} + \text{Other Pulses (1.79)} + \text{Tofu (3.16)} + \text{Wine (1.79)}) / 8 = 62.06 / 8 = 7.76$$

## References

- [1] IPCC (2022). Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Chapter 12.
- [2] FAO (2013). *Tackling Climate Change Through Livestock: A Global Assessment of Emissions and Mitigation Opportunities*. Food and Agriculture Organization of the United Nations, Rome.
- [3] Environment and Climate Change Canada (2025). *National Inventory Report: Greenhouse Gas Sources and Sinks in Canada, 1990–2023*.
- [4] Agriculture and Agri-Food Canada (2006). *An Overview of the Canadian Agriculture and Agri-Food System*. Table 9: Price Elasticities.
- [5] Deaton, A., & Muellbauer, J. (1980). An Almost Ideal Demand System. *American Economic Review*, 70(3), 312–326.
- [6] Poore, J., & Nemecek, T. (2018). Reducing food’s environmental impacts through producers and consumers. *Science*, 360(6392), 987–992.
- [7] Ritchie, H., Rosado, P., & Roser, M. (2022). Environmental Impacts of Food Production. *Our World in Data*. Based on Poore, J., & Nemecek, T. (2018). <https://ourworldindata.org/grapher/food-emissions-supply-chain>
- [8] Statistics Canada (2010). Table 11-10-0125-01: Detailed Food Spending, Canada, Regions and Provinces. <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1110012501>
- [9] Statistics Canada (2009). Food Statistics 2009. Catalogue no. 21-020-X.
- [10] Springmann, M., et al. (2025). A reform of value-added taxes on foods can have health, environmental and economic benefits in Europe. *Nature Food*.
- [11] Jensen, J. D., & Smed, S. (2013). The Danish tax on saturated fat: Short run effects on consumption, substitution patterns and consumer prices of fats. *Food Policy*, 42, 18–31.
- [12] Funke, F., et al. (2022). Toward Optimal Meat Pricing: Is It Time to Tax Meat Consumption? *Review of Environmental Economics and Policy*, 16(2).
- [13] Sureth, M., Plinke, C., et al. (2026). Environmental impacts from European food consumption can be reduced with carbon pricing or a value-added tax reform. *Nature Food*.

- [14] Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17, 261–272.
- [15] Anthropic (2025). Claude (Opus 4.6) [Large language model]. <https://www.anthropic.com>